SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 1 – Analytical Problem solving (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO2	**Assessment:**	Assessment 1: Analytical Problem solving
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO2 Statement:	***Apply image enhancement, filtering, and segmentation techniques to analyze images, detect edges and boundaries, and solve real-world Computer Vision problems.***		
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Section / Batch:	CSE	**Date:**	21-10-2026

Code of Conduct

I **Abburi Ganesh Ram Kamal 192311183** *that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.* Signature of the Student: _____

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Analytical Problems

Q1. A grayscale image contains the following pixel intensity values: 10, 10, 20, 20, 20, 30, 30, and 40. Construct the histogram by calculating the frequency of each intensity value. Calculate the cumulative distribution function (CDF) for the given image. Apply histogram equalization using the formula s = round((L − 1) × CDF), where L = 256. Determine the new intensity values and compare the contrast distribution before and after equalization.

A.

Given pixel values:

10, 10, 20, 20, 20, 30, 30, 40

Total number of pixels:

\[N=8 \]

1. Histogram

Intensity \(r_k\)	Frequency \(n_k\)	Probability \(P(r_k)=n_k/8\)	CDF
10	2	2/8 = 0.25	0.25
20	3	3/8 = 0.375	0.625
30	2	2/8 = 0.25	0.875
40	1	1/8 = 0.125	1.000

The **CDF** is calculated by accumulating the probabilities:

\[CDF(10)=0.25 \]\[CDF(20)=0.25+0.375=0.625 \]\[CDF(30)=0.625+0.25=0.875 \]\[CDF(40)=0.875+0.125=1 \]

2. Histogram Equalization

Given:

\[s=\operatorname{round}[(L-1)\times CDF] \]

where \(L=256\), so \(L-1=255\).

Original intensity	CDF	Equalized intensity \(s\)
10	0.25	\(round(255\times0.25)=\mathbf{64}\)
20	0.625	\(round(255\times0.625)=\mathbf{159}\)
30	0.875	\(round(255\times0.875)=\mathbf{223}\)
40	1.000	\(round(255\times1)=\mathbf{255}\)

Therefore, the transformed image values are:

\[\boxed{64,64,159,159,159,223,223,255} \]

3. Comparison

Before equalization: pixels are concentrated in the narrow range **10–40**.

After equalization: pixels are spread across approximately **64–255**.

Thus, histogram equalization **increases the dynamic range and improves contrast**, making differences between intensity levels more pronounced.

Q2. A 3 × 3 image has the intensity values [[10, 20, 30], [20, 100, 20], [10, 20, 30]]. Apply a 3 × 3 averaging filter to calculate the new intensity value of the centre pixel. Replace the centre pixel with the calculated average while keeping the other pixels unchanged. Apply a median filter to the same neighbourhood and calculate the centre pixel value. Compare the results of the averaging and median filters and determine which filter better preserves image details in the presence of noise.

A.

Given:

$$\begin{bmatrix} 10&20&30\\ 20&100&20\\ 10&20&30 \end{bmatrix}$$

The centre pixel is **100**, which can be considered a noisy/high-intensity pixel.

1. 3 × 3 Averaging Filter

Add all 9 pixels:

$$10+20+30+20+100+20+10+20+30=260$$

Average:

$$\frac{260}{9}=28.89$$

Assuming integer intensity:

$$\boxed{29}$$

So the filtered image becomes approximately:

$$\begin{bmatrix} 10&20&30\\ 20&29&20\\ 10&20&30 \end{bmatrix}$$

2. Median Filter

Arrange all values in ascending order:

$$10,10,20,20,20,20,30,30,100$$

There are 9 values, so the middle (5th) value is:

$$\boxed{20}$$

Thus, the centre pixel becomes **20**.

Comparison

- **Averaging filter:** centre = **29**
- **Median filter:** centre = **20**

The value **100** is an outlier. The averaging filter is strongly influenced by this value, whereas the median filter completely rejects its influence.

Therefore:

$$\boxed{\text{Median filter better preserves details in the presence of impulse noise}}$$

It is particularly effective against **salt-and-pepper noise**

Q3. Consider a grayscale image with the intensity matrix [[50, 50, 50], [50, 100, 50], [50, 50, 50]]. Apply the Laplacian filter mask [[0, −1, 0], [−1, 4, −1], [0, −1, 0]] to calculate the response at the centre pixel. Use the calculated response to sharpen the centre pixel by adding it to the original intensity value. Determine the resulting centre pixel intensity after sharpening. Explain how the Laplacian filter enhances intensity transitions and fine details in the image.

A. Given:

$$\begin{bmatrix} 50&50&50\\ 50&100&50\\ 50&50&50 \end{bmatrix}$$

Laplacian mask:

$$\begin{bmatrix} 0&-1&0\\ -1&4&-1\\ 0&-1&0 \end{bmatrix}$$

1. Calculate the Laplacian response

Only the centre and its four direct neighbours contribute:

$$R=(4\times100)-(50+50+50+50)$$$$R=400-200$$$$\boxed{R=200}$$

2. Sharpen the centre pixel

The question specifies adding the response to the original pixel:

$$g(x,y)=f(x,y)+R$$$$=100+200$$$$\boxed{300}$$

So the resulting centre intensity is **300**.

In a real 8-bit image, intensity values are normally clipped to 255, so it would become **255**. Under the formula given in the question, the calculated value is **300**.

Explanation

The Laplacian detects rapid intensity changes. Here, the centre pixel (**100**) is much brighter than its surrounding pixels (**50**), producing a large Laplacian response.

Adding this response emphasizes the intensity transition and therefore **sharpens edges and fine details**.

Q4. A grayscale image contains the pixel intensity values [[40, 60, 120], [80, 150, 200], [30, 100, 180]]. Apply global thresholding with a threshold value of 100, assigning 1 to pixels greater than or equal to the threshold and 0 to the remaining pixels. Construct the resulting binary image. Calculate the number of foreground and background pixels after thresholding. Explain how the selected threshold separates objects from the background and identify one limitation of this method.

A.

Given:

$$\begin{bmatrix} 40&60&120\\ 80&150&200\\ 30&100&180 \end{bmatrix}$$

Threshold:

$$T=100$$

Rule:

$$g(x,y)= \begin{cases} 1,&f(x,y)\ge100\\ 0,&f(x,y)<100 \end{cases}$$

Apply the rule:

$$\boxed{ \begin{bmatrix} 0&0&1\\ 0&1&1\\ 0&1&1 \end{bmatrix}}$$

Foreground pixels

Values ≥ 100:

$$ 120,150,200,100,180 $$

Number of foreground pixels:

$$ \boxed{5} $$

Background pixels

Values < 100:

$$ 40,60,80,30 $$

Number of background pixels:

$$ \boxed{4} $$

Explanation

The threshold 100 separates the image into two classes:

- **0:** background, intensity below 100
- **1:** foreground/object, intensity 100 or greater

This is useful when the object and background have sufficiently different intensity ranges.

Limitation: global thresholding uses one threshold for the entire image. If lighting is uneven or the object and background have overlapping intensities, segmentation can be poor.

Q5. A grayscale image has the intensity matrix [[10, 10, 10], [10, 50, 90], [10, 50, 90]]. Apply the Sobel operator using the masks Gx = [[−1, 0, 1], [−2, 0, 2], [−1, 0, 1]] and Gy = [[−1, −2, −1], [0, 0, 0], [1, 2, 1]] to calculate the gradient components at the centre pixel. Calculate the gradient magnitude using $G = \sqrt{(Gx^2 + Gy^2)}$. Determine the dominant edge direction based on the calculated gradient components. Explain how the detected edge information can be used for boundary detection and image segmentation.

A.
Given:

$$ I= \begin{bmatrix} 10&10&10\\ 10&50&90\\ 10&50&90 \end{bmatrix} $$

Sobel masks:

$$ G_x= \begin{bmatrix} -1&0&1\\ -2&0&2\\ -1&0&1 \end{bmatrix} $$ $$ G_y= \begin{bmatrix} -1&-2&-1\\ 0&0&0\\ 1&2&1 \end{bmatrix} $$

1. Calculate G_x

Multiply corresponding values:

$$ G_x=(-1)(10)+(0)(10)+(1)(10) $$ $$ +(-2)(10)+(0)(50)+(2)(90) $$ $$ +(-1)(10)+(0)(50)+(1)(90) $$ $$ =-10+10-20+180-10+90 $$ $$ \boxed{G_x=240} $$

2. Calculate G_y

$$ G_y=(-1)(10)+(-2)(10)+(-1)(10) $$ $$ +(0)(10)+(0)(50)+(0)(90) $$ $$ +(1)(10)+(2)(50)+(1)(90) $$ $$ =-10-20-

10+10+100+90 \]\[\boxed{G_y=160} \]

3. Gradient magnitude

\[G=\sqrt{G_x^2+G_y^2} \]\[G=\sqrt{240^2+160^2} \]\[=\sqrt{57600+25600} \]\[=\sqrt{83200} \]\[\boxed{G\approx288.44} \]

4. Dominant edge direction

Since:

\[|G_x|=240 > |G_y|=160 \]

the gradient has a stronger **horizontal-gradient component**.

The gradient orientation is:

\[\theta=\tan^{-1}\left(\frac{G_y}{G_x}\right) \]\[\theta=\tan^{-1}\left(\frac{160}{240}\right) \]\[\boxed{\theta\approx33.7^\circ} \]

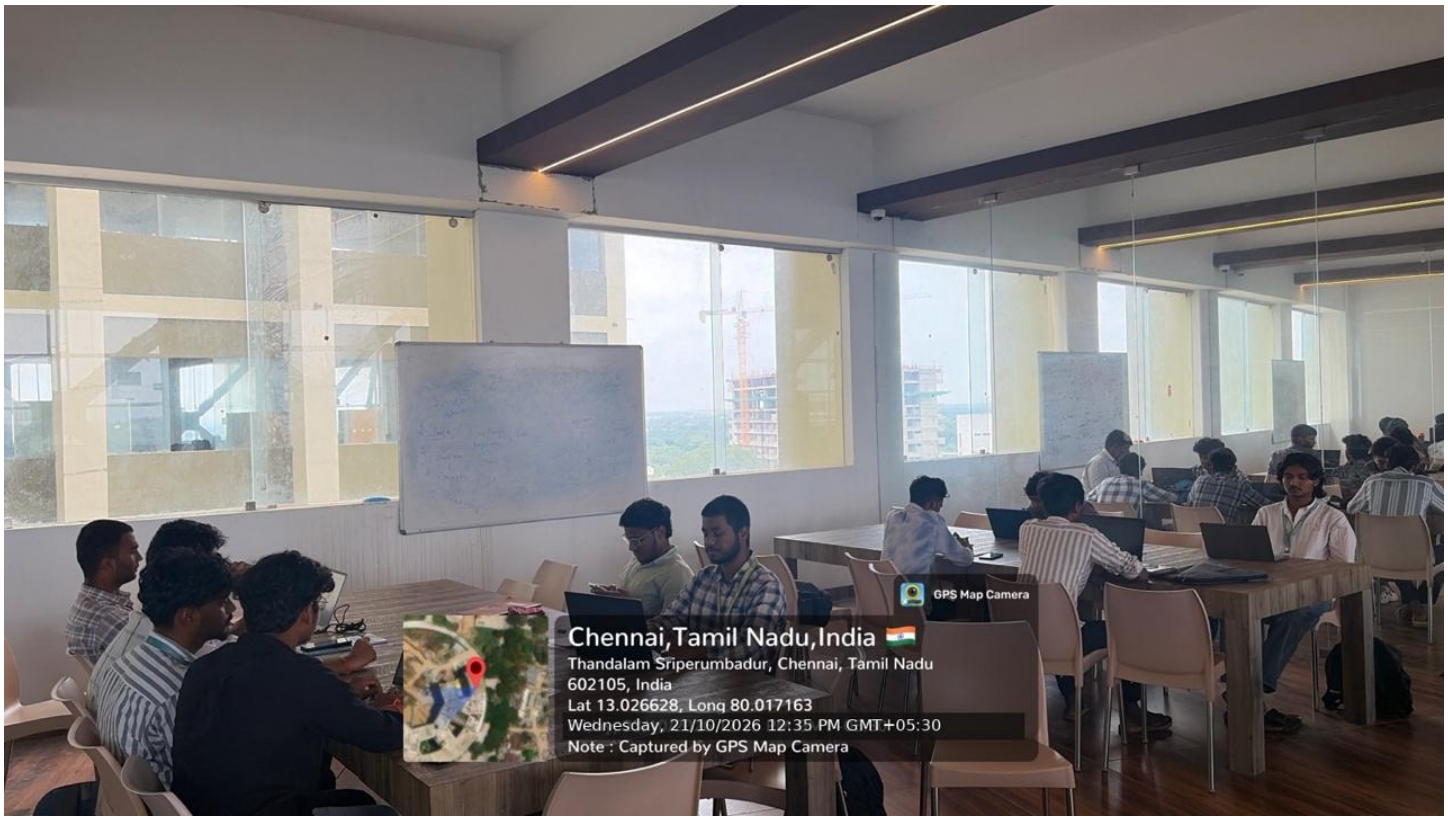
So the gradient direction is approximately **33.7°** from the positive x-axis. The corresponding edge itself is perpendicular to the gradient, approximately:

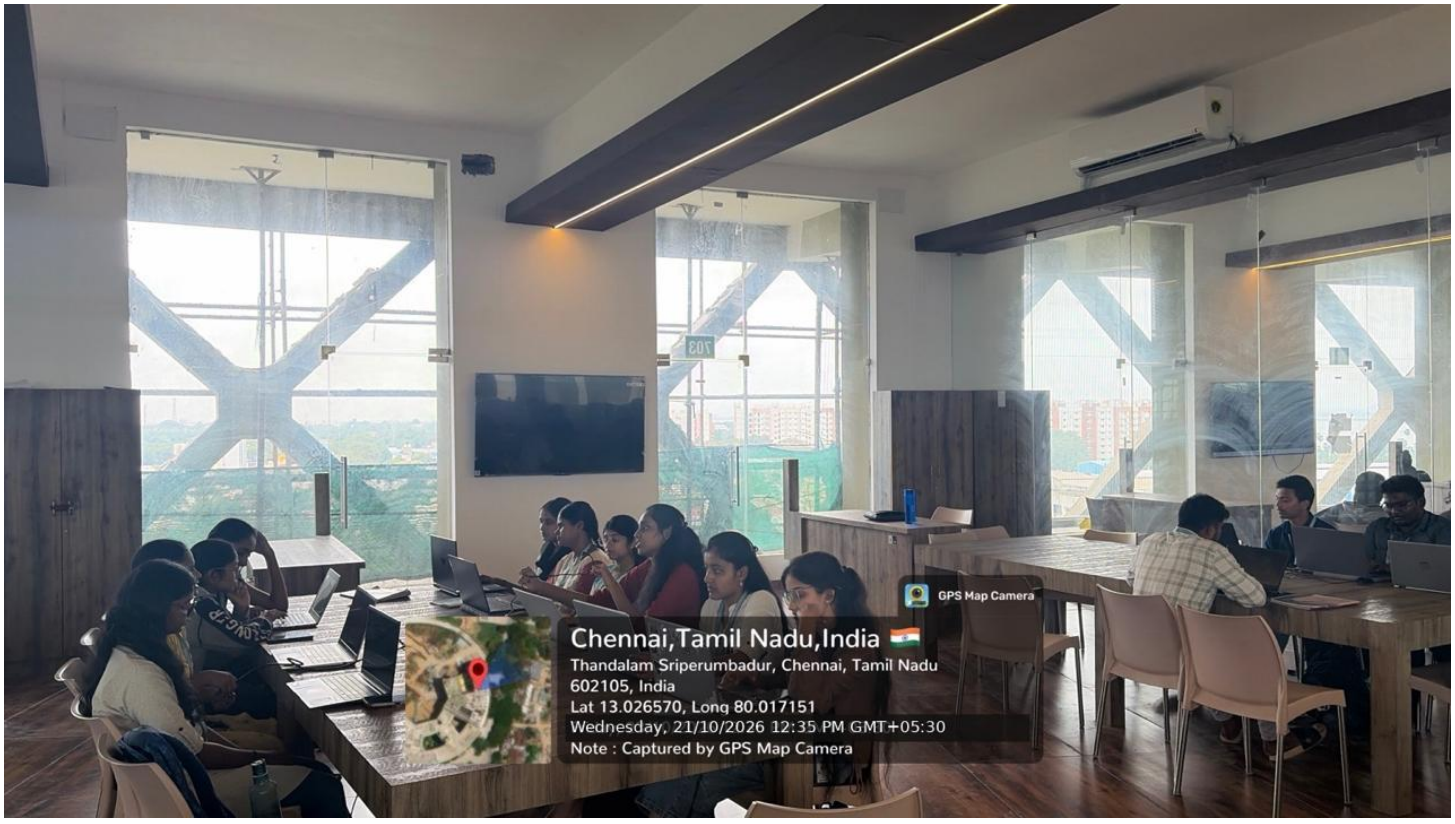
\[\boxed{123.7^\circ} \]

Application

The Sobel operator identifies locations where intensity changes rapidly. These strong gradient values can be used to:

- detect **object boundaries**,
- locate **edges and contours**,
- separate regions during **image segmentation**, and
- identify shapes and structures in an image.





The following Assessment has been discussed by the Faculty during class on **21-10-2026**

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
cuss	Signature: _____ Date: _____	Signature: _____ Date: _____